

Chapter

Quadratic Equation In One Variable

1. The roots of the quadratic equation $px^2 - qx + r = 0$ are real and equal if:

(a) $p^2 = 4qr$

(b) $q^2 = 4pr$

(c) $-q^2 = 4pr$

(d) $p^2 > 4qr$

Main 2024

2. If the roots of equation $x^2 - 6x + k = 0$ are real and distinct, then value of k is:

(a) > -9

(b) > -6

(c) < 6

(d) < 9

SQP 2024

3. If 3 is a root of the quadratic equation $x^2 - px + 3 = 0$ then p is equal to :

(a) 4

(b) 3

(c) 5

(d) 2

Main 2023

4. Find the value(s) of k if the quadratic equation $3x^2 - k\sqrt{3}x + 4 = 0$ has real roots.

Comp 2023

5. A man covers a distance of 100 km, travelling with a uniform speed of x km/hr. Had the speed been 5 km/hr more it would have taken 1 hour less. Find x the original speed.

Main 2023

6. The roots of $100x^2 - 20x + 1 = 0$ is :

(a) $\frac{1}{20}$ and $\frac{1}{20}$

(b) $\frac{1}{10}$ and $\frac{1}{20}$

(c) $\frac{1}{10}$ and $\frac{1}{10}$

(d) none of the above

Main 2022 Sem I

7. The roots of the quadratic equation $4x^2 - 7x + 2 = 0$ are 1.390 and 0.359.

The roots correct to 2 significant digits are :

(a) 1.39 and 0.36

(b) 1.3 and 0.35

(c) 1.4 and 0.36

(d) 1.390 and 0.360

Main 2022 Sem I

8. Which of the following quadratic equation has 2 and 3 as its roots?

(a) $x^2 - 5x + 6 = 0$

(b) $x^2 + 5x + 6 = 0$

(c) $x^2 - 5x - 6 = 0$

(d) $x^2 + 5x - 6 = 0$

Main 2022 Sem I

9. Solve the following quadratic equation:

$$x^2 + 4x - 8 = 0$$

Give your answer correct to one decimal place. (Use mathematical tables if necessary.)

Main 2023

10. Two taps running together can fill a cistern in $2\frac{8}{11}$ minutes. If one tap takes 1 minute more than the other to fill the cistern, find the time in which each tap separately can fill the cistern.

Comp 2023

Solution

1.

Ans: For the quadratic equation

$px^2 - qx + r = 0$, the roots are **real and equal** if the discriminant is zero.

$$b^2 - 4ac = 0$$

Here, $a = p$, $b = -q$, $c = r$

$$(-q)^2 = 4pr$$

$$q^2 = 4pr$$

Hence, **option (b) $q^2 = 4pr$** is correct.

2.

Ans: (d) $k < 9$

Explanation:

For a quadratic equation to have real and distinct roots, the discriminant must be greater than zero.

Here, for $x^2 - 6x + k = 0$:

$$b^2 > 4ac \Rightarrow 36 > 4k \Rightarrow k < 9.$$

3.

Ans: (a) 4

Explanation:

Since 3 is a root of the equation $x^2 - px + 3 = 0$, substitute $x = 3$:

$$3^2 - 3p + 3 = 0 \Rightarrow 12 - 3p = 0 \Rightarrow p = 4$$

Hence, the correct answer is **4**.

4.

Ans: For the quadratic equation $3x^2 - k\sqrt{3}x + 4 = 0$ to have real roots, the discriminant must be greater than or equal to zero.

Here,

$$a = 3, b = -k\sqrt{3}, c = 4$$

So,

$$b^2 - 4ac \geq 0$$

$$(-k\sqrt{3})^2 - 4(3)(4) \geq 0$$

$$3k^2 - 48 \geq 0$$

$$k^2 \geq 16$$

Hence,

$$k \leq -4 \text{ or } k \geq 4$$

So, the required values of k are $k \leq -4$ or $k \geq 4$.

5.

Ans:

Let the original speed of the man be x km/hr.

$$\text{Time taken at speed } x = \frac{100}{x} \text{ hours}$$

If the speed were $x + 5$ km/hr, time taken would be:

$$\frac{100}{x + 5} \text{ hours}$$

Given that the increased speed takes **1 hour less**:

$$\frac{100}{x + 5} = \frac{100}{x} - 1$$

Solving this equation gives:

$$x = 20$$

(The negative value is rejected as speed cannot be negative.)

$$\text{Original speed of the man} = 20 \text{ km/hr}$$

6.

Ans: (c) $\frac{1}{10}$ and $\frac{1}{10}$

Explanation:

The given quadratic equation is

$$100x^2 - 20x + 1 = 0$$

On factorising,

$$(10x - 1)^2 = 0$$

$$x = \frac{1}{10}, \frac{1}{10}$$

Hence, the roots are equal and both are $\frac{1}{10}$.

7.

Ans: (c) 1.4 and 0.36

Explanation:

To round the given roots to **two significant digits**, we follow the rules of significant figures.

- For **1.390**, the first two significant digits are **1** and **4** (since the next digit is 9, it rounds up).

Hence, **1.390** $\approx$ **1.4** (to two significant figures).

- For **0.359**, the first two significant digits are **3** and **6** (since the next digit is 9, it rounds up).

Hence, **0.359** $\approx$ **0.36** (to two significant figures).

Therefore, the correct answer is **1.4 and 0.36**, i.e. **option (c)**.

8.

Ans: (a) $x^2 - 5x + 6 = 0$

Explanation:

If 2 and 3 are the roots of a quadratic equation, then the factors of the equation are $(x - 2)$ and $(x - 3)$.

$$(x - 2)(x - 3) = 0$$

Expanding,

$$x^2 - 3x - 2x + 6 = 0$$

$$x^2 - 5x + 6 = 0$$

Hence, the correct answer is **(a)**.

9.

Ans:

Given

$$x^2 + 4x - 8 = 0$$

Using the quadratic formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{Here, } a = 1, b = 4, c = -8$$

$$x = \frac{-4 \pm \sqrt{16 + 32}}{2} = \frac{-4 \pm \sqrt{48}}{2}$$

$$\sqrt{48} = 4\sqrt{3} \approx 6.92$$

$$x = \frac{-4 + 6.92}{2} = 1.46$$

$$x = \frac{-4 - 6.92}{2} = -5.46$$

Hence, the solutions correct to one decimal place are:

$$x = 1.5 \text{ and } x = -5.5$$

10.

Ans:

The two taps together fill the cistern in $2\frac{8}{11}$ minutes, that is $\frac{30}{11}$ minutes.

Hence, together they fill $\frac{11}{30}$ of the cistern in one minute.

Let the first tap fill the cistern in x minutes.

Then the second tap fills it in $x + 1$ minutes.

$$\frac{1}{x} + \frac{1}{x+1} = \frac{11}{30}$$

Solving this equation, we get:

$$x = 5$$

(The negative value obtained is rejected.)

Therefore, **the first tap fills the cistern in 5 minutes and the second tap fills it in 6 minutes.**